

Replacement Engine Control Unit

Complete documentation: the machine, the circuits, the specification, and the research all of it rests on.

This replaces the **Engine Control Unit** block on page 1 of the Kirloskar troubleshooting wiring diagram, manual 08-GP3-60-001. Every pin, signal and colour in these pages was read out of that document or measured in simulation; where something is inferred rather than confirmed, it says so on the line.

The engine has not been inspected yet. Twelve items remain open in the unknowns register, and the CPCB IV+ certification position in Memo 06 is a decision for the machine's owners, not an engineering conclusion. **Read that memo before fitting anything.**

CONTENTS

1. Field Handbook

The machine, its interface, the six circuits, and how to run it

2. Sensor Reference

Every sensor: what it measures, how we read it, what we do with it

3. Reconciled Specification

Binding source of truth -- pin map, power, deviations, unknowns

4. Memo 01 -- Engine Identity

Which engine this actually is, established from five hardware constraints

5. Memo 02 -- KG640C Controller

What the existing genset controller does, and what it needs from us

6. Memo 03 -- ECU Connector

The 94-pin connector: candidates, sourcing, and what stays unconfirmed

7. Memo 04 -- Injector Drive

Bank sharing proved safe, and the drive envelope it has to hit

8. Memo 05 -- MCU Selection

Why the S32K148, and the timing argument that actually binds

9. Memo 06 -- Standards & Law

ISO, CISPR and the CPCB IV+ certification position

10. Memo 07 -- BOM & Sourcing

Costed bill of materials, India sourcing, single-source flags

11. Memo 08 -- Engine Inspection Brief

The measurements to take at the machine, grouped by machine state

12. Phase 1 Research Plan

The plan the eight memos were written against